

Streamoid: Improving business productivity in fashion retail with a seamless cataloging experience

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Streamoid 

About Streamoid

Streamoid helps
retailers all around
the world drive better
by enabling a fast
seamless cataloging
experience with its
platform support

Streamoid is a fashion artificial intelligence (AI) innovator with a vision to power intelligent shopping experiences for shoppers and retailers across the globe. Whether it's online or through brick and mortar stores, Streamoid offers solutions that help businesses improve their gross margins. Based out of Bangalore and New York, the startup has customers in countries all around the world.

Industries: Retail & Consumer Goods

Google Kubernetes and Cloud Run.

Location: India

Google Cloud results

- Consistently upgrades more than 150 ML models automatically to allow more operational flexibility
- Reduces DevOps time from 16 hours a week to two hours per week with Google Cloud automation
- Saves up to 80% of infrastructure cost with pay-as-you-go model

Helping
businesses
increase
cataloging
speed by 10x

Google Cloud (<https://cloud.google.com/>)

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Cloud Run (<https://cloud.google.com/run/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

- Enables
global
market
growth
without
worrying
about data
compliance

The fashion industry is growing at a speed like never before, particularly in the Asia-Pacific and European regions. By 2030, it is expected to become a US\$3 trillion industry.

(<https://www.statista.com/topics/965/apparel-market-in-the-us/#:~:text=In%202019%2C%20global%20retail%20sales,Asia%2DPacific%20and%20European%20regions.>)

. To keep up with this fast pace, Streamoid, a fashion AI company, was founded to help retailers drive better margins by streamlining their processes using data as a decision-making tool.

The company has global brands such as Forever 21 and American Outfitters under its belt, alongside local enterprises in different regions of the world.

"Retailers are introducing a lot of new technologies. From ecommerce websites to digitizing their brick and mortar stores, we offer a range of applications to help them boost sales and revenue, regardless of the platform," says Vivek Bharadwaj, Director, Product Management at Streamoid.

"We currently have more than 150 machine learning (ML) models and are adding more as the complexity of our product grows and we add more features for customers... With Google Kubernetes Engine and Cloud Run, each ML model can operate independently, allowing greater flexibility for the team."

—Vivek Bharadwaj,
Director, Product
Management, Streamoid

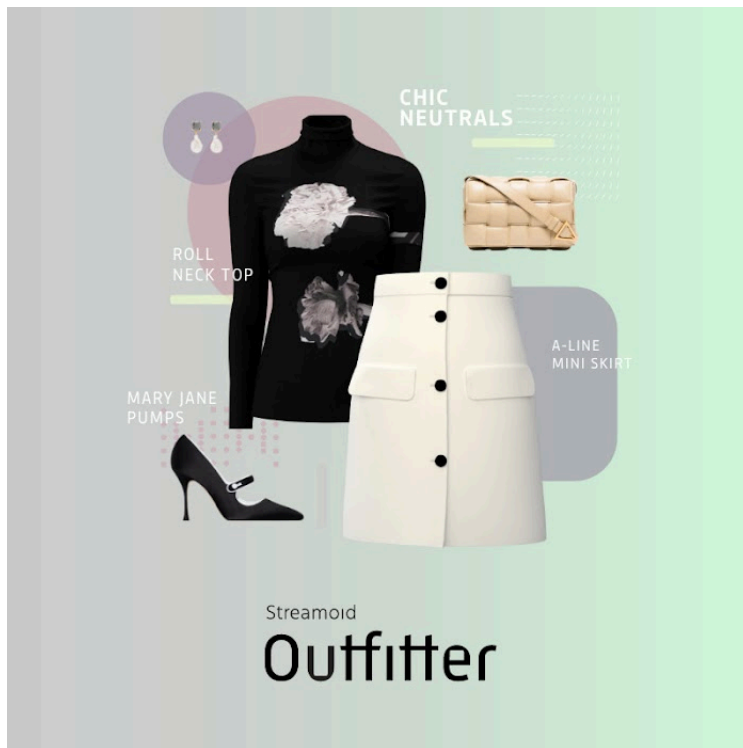
As a whole, the company offers a variety of services that range from a recommendation system to search offering tools. But its key product is Autoscribe, a smart tagging and content generation tool that helps brands catalog and streamline their products. Essentially, brands provide the images and Autoscribe generates the appropriate tags and descriptions for each product. It also ensures that the images are accurately categorized and optimized for discovery.

"From runway to street trends, fashion is an image-driven industry," says Bharadwaj. "This is why we felt that having a strong product that combines computer vision and AI technology was imperative, and we decided to focus on AI as our core technology to propel Autoscribe over the past three years."

To help stabilize and boost Autoscribe's AI capabilities for growth, Streamoid looked to Google Cloud (<https://cloud.google.com/>) in January 2021 to speed up its cataloging process. It found early success with Google Kubernetes Engine (/kubernetes-engine) (GKE) and Cloud Run (/run) for its containerization capabilities that support Autoscribe's capabilities and propel it for growth.

"We currently have more than 150 machine learning (ML) models and are adding more as the complexity of our product grows and we add more features for customers," says Bharadwaj. Each of Streamoid's ML models needs to be continuously upgraded. If they're packaged together, then every time they need to update a model the DevOps would need to

unpack and repack everything back, a laborious and time-consuming process. "With Google Kubernetes Engine and Cloud Run, each ML model can operate independently, allowing greater flexibility for the team."



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Supporting major global apparel brands with efficient AI cataloging

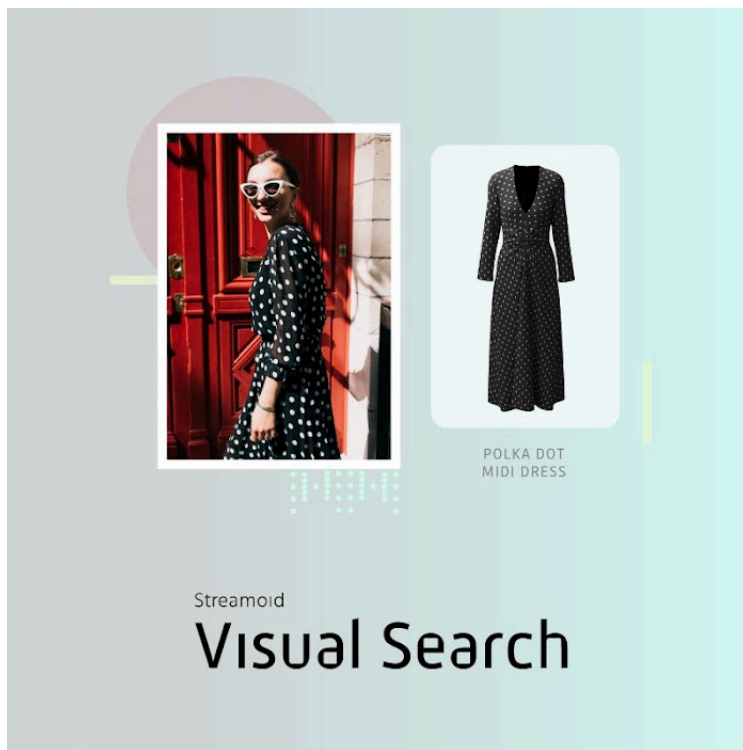
The fast-paced nature of fashion retail brings along with it some unique challenges. These include keeping up with the ever-increasing number of marketplaces for brands as they grow, and the need to train and retrain employees as companies and the technology alongside them evolve.

Before Google Cloud, the company used a hybrid setup, hosting its ML models in-house and using virtual machines (VMs) from different cloud providers to host its web applications. Just ensuring

that everything runs smoothly on a day-to-day basis was very time-consuming and tedious given the turnaround of new products and requirements. Now, Streamoid's entire cataloging solution is hosted on Google Cloud, significantly reducing the amount of time and energy required to train the various VM models.

"We wanted to have a setup that involved minimal to no DevOps input after deployment, because at the end of the day we are a startup with limited resources," explains Bharadwaj. "Cloud Run was the perfect solution for us as it takes one click on a button and everything is taken care of." Since deploying Cloud Run, Streamoid has reduced DevOps time from 16 hours a week to just two hours a week. At the same time, the issue resolution time has been cut from eight hours to just one to two hours.

The inconsistent nature of cataloging is why having a scalable platform was critical to the business. While on some days there may be a lot of traffic, there are days where the volume of products being processed is low. Manual scaling to handle the rapidly increasing cataloging requirements is not only complicated, but costly because of the nature of GPUs (graphics processing units). However, with Google Cloud autoscaling and the pay-as-you-go model offered by Cloud Run, Streamoid is able to save up to 70%–80% of its infrastructure cost.



Ensuring ML and data compliance around the world for faster growth

From Europe to the US, even Australia, Streamoid already has customers all around the world and aims to continue growing its market presence. Google Cloud helps ensure that it is able to expand quickly and seamlessly, since all of its ML models can be colocated, reducing its compliance burden. Different customers have preferences on where they want their data to be hosted, which can be challenging to meet with in-house data centers. With a global presence, Google Cloud automatically tackles the issue of compliance.

"With colocation, all our ML models are easily transferable to wherever our customers are, and this has made the process of just moving data across markets really easy," says Bharadwaj.

At the same time, Google Cloud ensures lower latency. This means large volumes of data can be transferred at any time, across continents without delay. Bharadwaj explains that in the cataloging business, the more images and products a company can manage, the bigger the business can grow. "Ultimately, it's a data race. The faster data can reach us, the more data we are able to process, which means more opportunities for businesses to scale." With GKE and Cloud Run, Streamoid has already seen a 10x growth. Now, it has ambitious plans to grow a further 10x, reaching its target of 50 million products, without adding more headcount to its infrastructure team.

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Exploring more Google Cloud capabilities for better speed and agility

The Streamoid team is currently exploring [Datastore](#) (/datastore) to expand its data storage capabilities and [Cloud TPU](#) (/tpu) to train new ML models in the cloud, so it can further speed up the process. The team will also be looking into [API Gateway](#) (/api-gateway) to better manage all of its APIs in a more systematic manner.

Bharadwaj says, "Today, we are still limited by our hardware in terms of the number of ML models we can generate, but I believe that Cloud TPU is our solution to scale and serve even more customers on a greater scale."

